

# Noise-Aware ANNNI

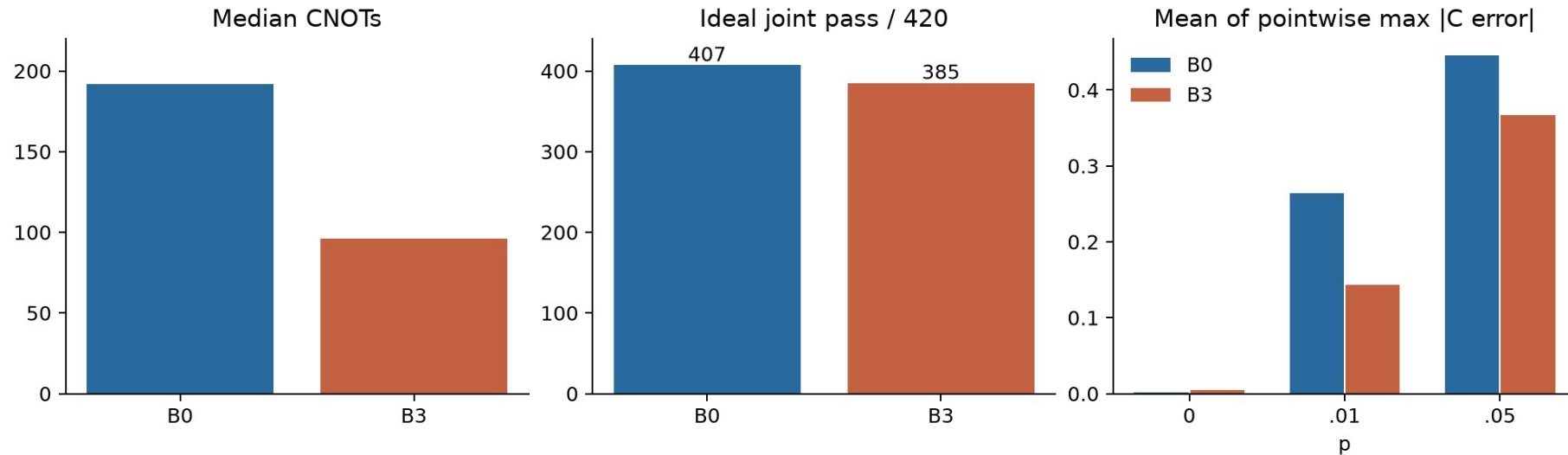
## Phase Diagnostics

Low-CNOT preparation, error mitigation, and their limits

Q-SITE 2026 / Scientific Track

Team hamster

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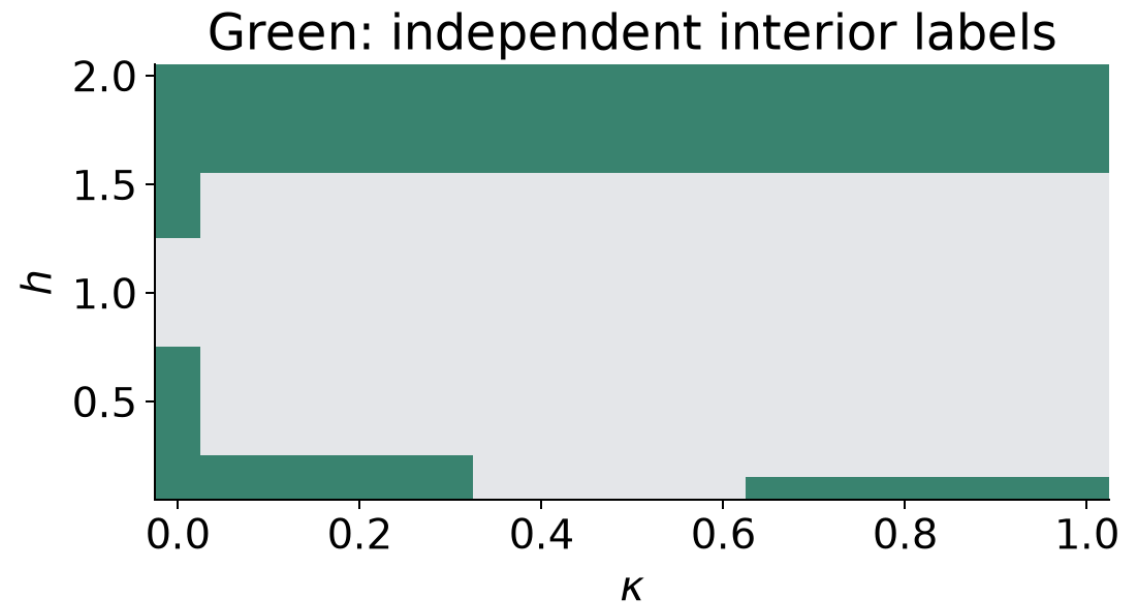
# Model and evidence layers

Periodic N8/N12 circuits; large-size OBC evidence kept separate

$H = -\sum ZZ(nn) + \kappa \sum ZZ(nnn) - h \sum X$

Every CNOT: target-only depolarizing channel

R0 interiors / RN finite-size features / Rlarge trends



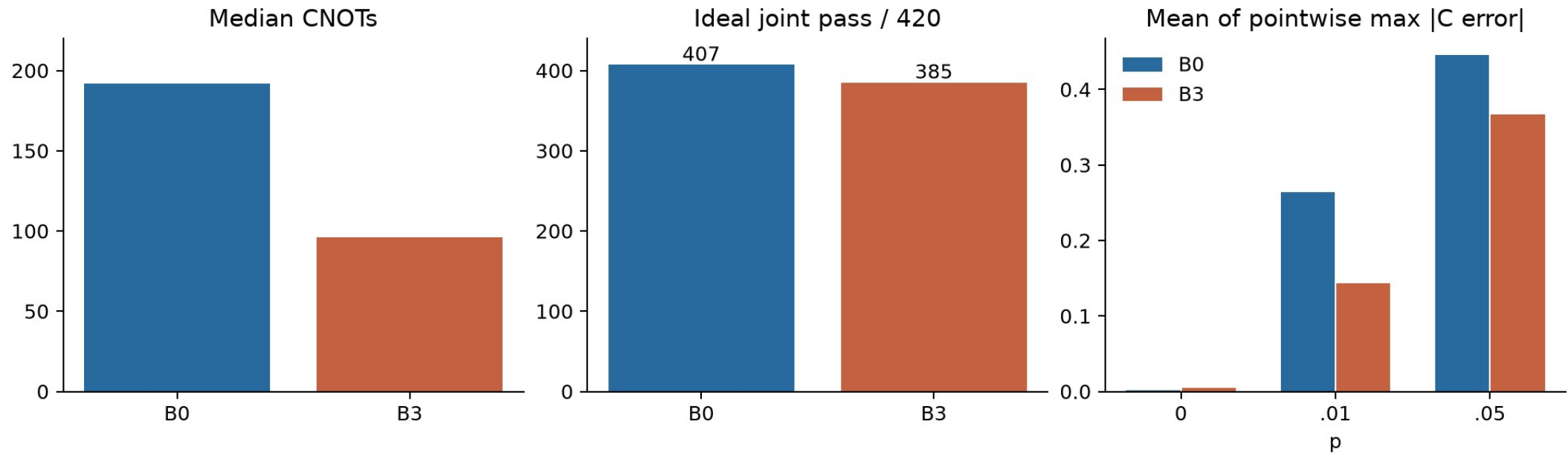
# Fewer gates, different tradeoffs

B0 fixed six-layer HVA -> B3 multi-reference adaptive preparation

Median CNOTs: 192 -> 96

Ideal joint passes: 407/420 -> 385/420

$p=.01$  correlation error improves at 402/420 points



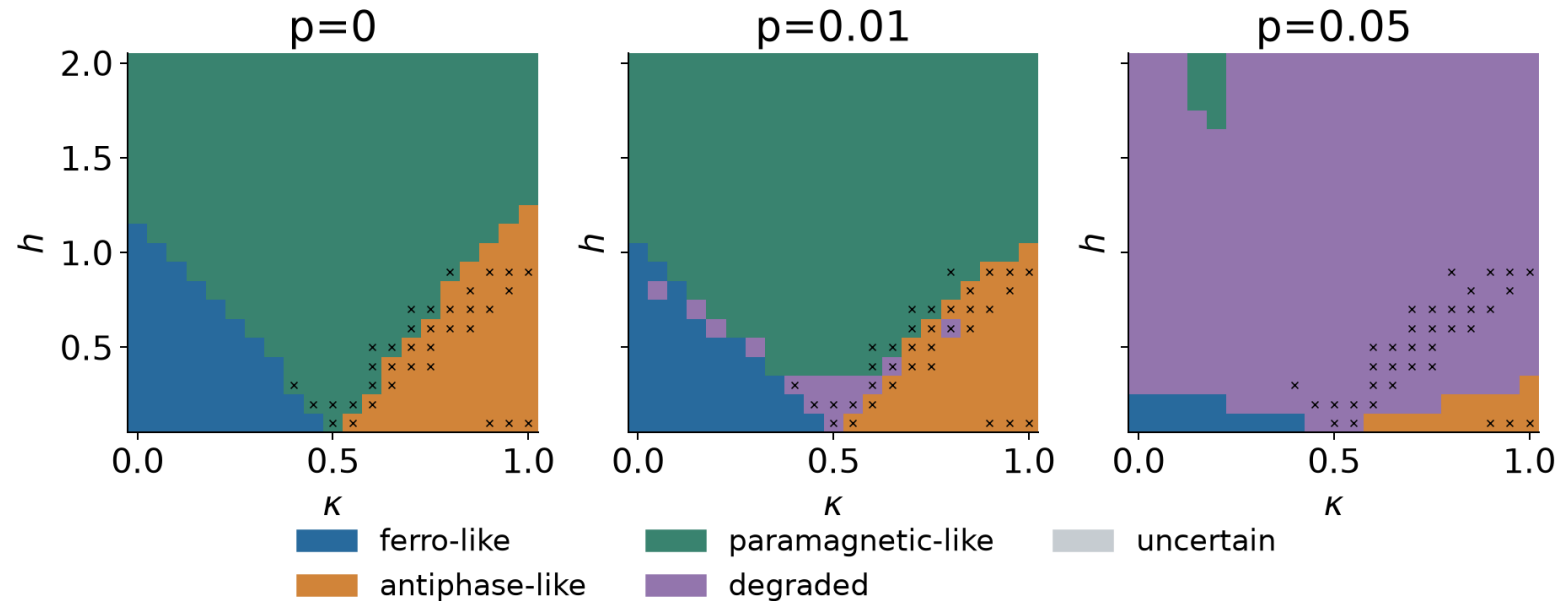
# Three noise levels, one diagnostic

Frozen D3; exact expectations; identical 420-point grid

Colors: estimated phase-like regions

Crosses: ideal preparation failed

Uncertain and degraded remain distinct



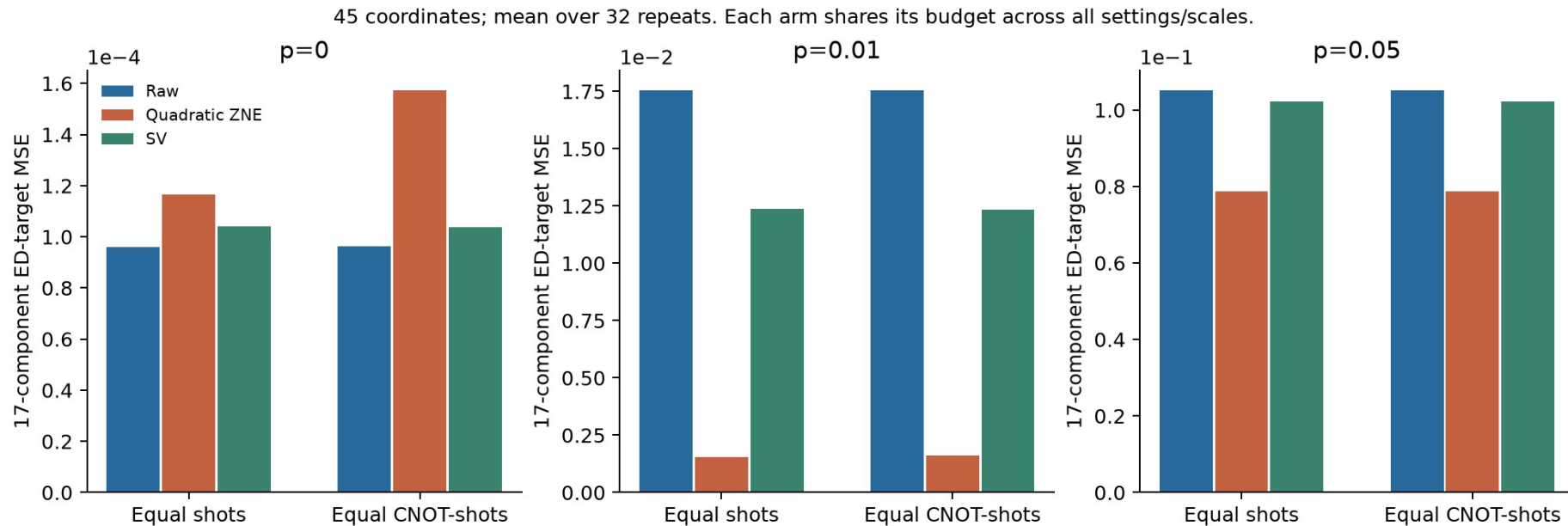
# Mitigation at equal CNOT-shots

45 fixed coordinates / 32 repeats / unchanged 17-component MSE

$G = 100,000 \times \text{original CNOT count}$

Raw / SV: 100k shots; ZNE: 33,334

ZNE MSE decrease: 90.75% / 25.19%



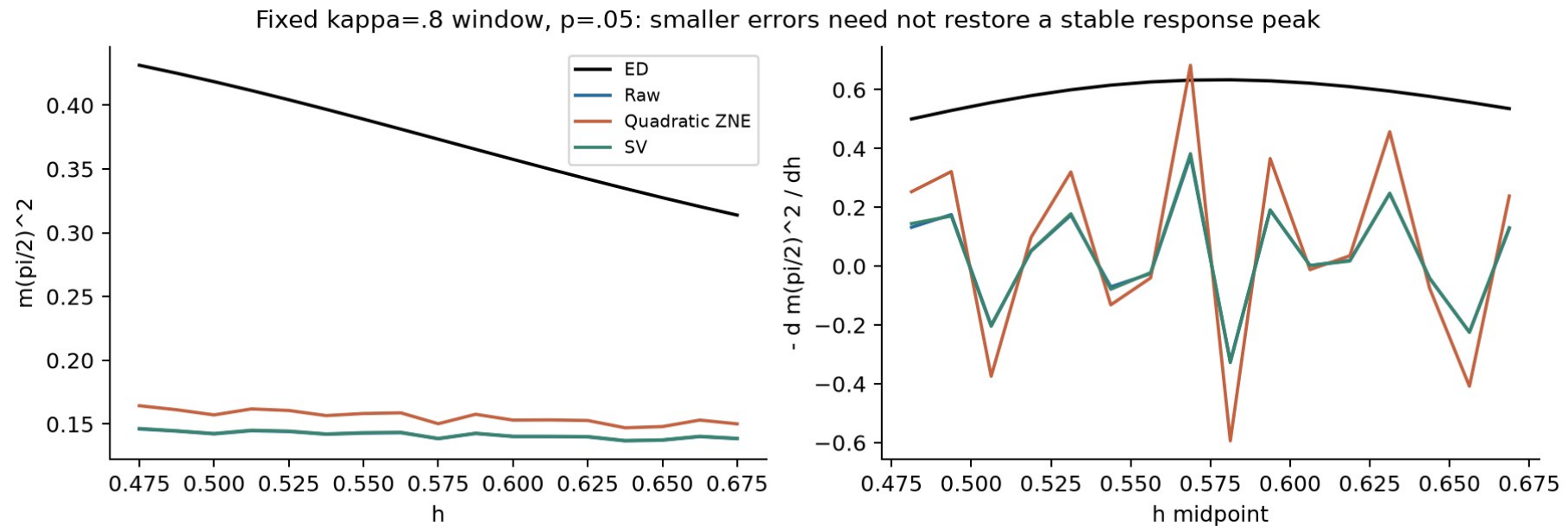
# Better observables, unstable features

Six complete 17-point windows; only five resolvable references

$\kappa=.5$  remains reference-unresolved

Endpoints, multiple peaks and branch effects retained

Matching is finite-size evidence, not a phase certificate



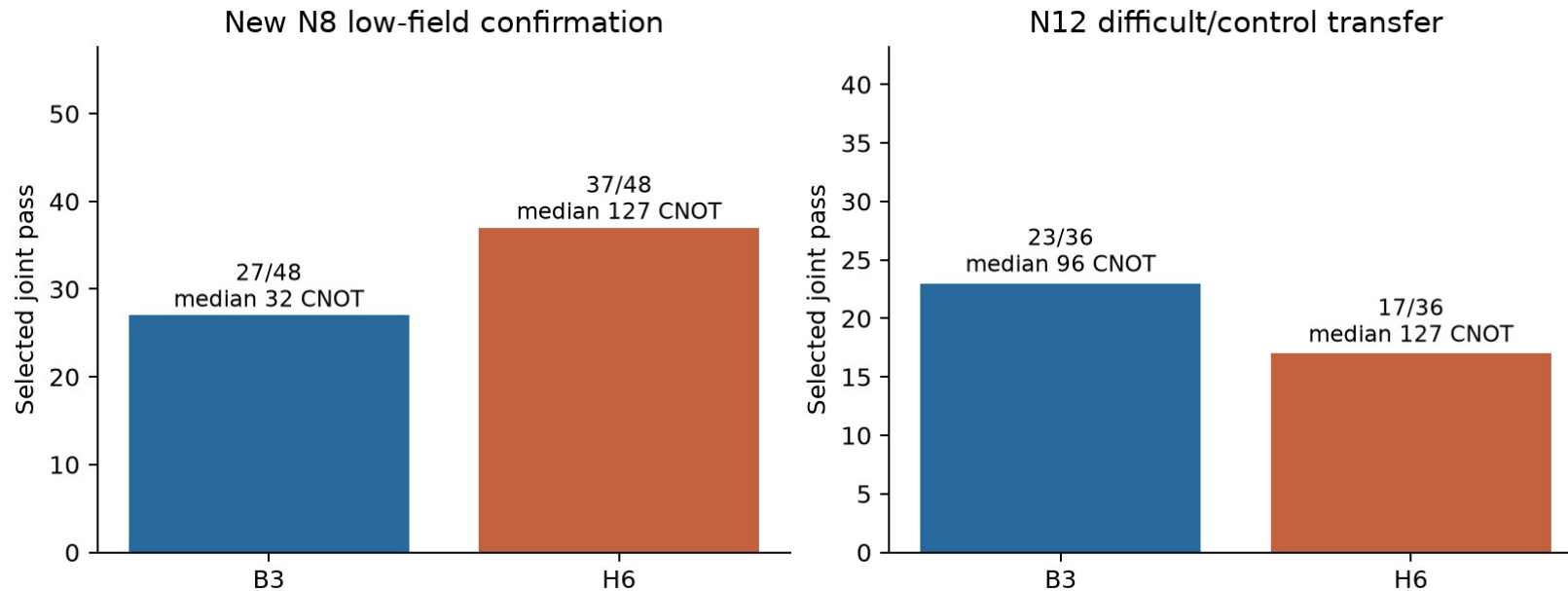
# Extensions expose the limits

H6 is a physical/domain-wall union, not the six-layer HVA

N8 low-field passes: 27/48 -> 37/48, with more resources

N12 difficult/control transfer: 23/36 vs 17/36

Dynamics, multiple detectors and mitigation retained



# What the evidence supports

A reproducible observable result with explicit phase-diagram limits

Lower-CNOT preparation helps noisy correlations

Equal-resource ZNE reduces bias, not every failure

No supported continuous floating interval

